

Challenges in Running AI on Embedded Computers



This is an extract from a speech presented by Nikhil Bhaskaran, founder, Shunya OS, at [iotshow.in](https://www.iotshow.in) in 2019.

Shunya OS is an operating system that optimises AI libraries for embedded systems, to enable developers to build AI on edge solutions very fast and at a very low cost. In the next five years, AI is going to be inside everything; and edge analytics is going to be the next big thing for AI.

The work going on in the artificial intelligence (AI) space nowadays is more on the application side to solve industry problems. And people want to get the solution into the market as fast as possible. For this, most of the AI libraries are built into the cloud. In that, there is a large stack, where the first challenge starts with data.

You need to train a model. It is not creating a model that is the biggest challenge, but finding data and arranging it in a way to efficiently train your model is. So, there is a large market for getting the right data.

The second part is building a model. There are many applications that do not require creating a model like face

recognition and object detection. Because these common applications already have ready models available out there—one just needs to choose the right one for the application. And after that, you need to further train your model, as most models are usually trained for different sets of data which do not give a good output.

As each and every library is good at one thing or the other, one needs to carefully choose the right one that is suitable for the application.

After having a working model, it must be used purposefully to generate enough money. Some of the questions which the entrepreneurs should focus on are: What should it (the model) be used for? What is the market looking for?